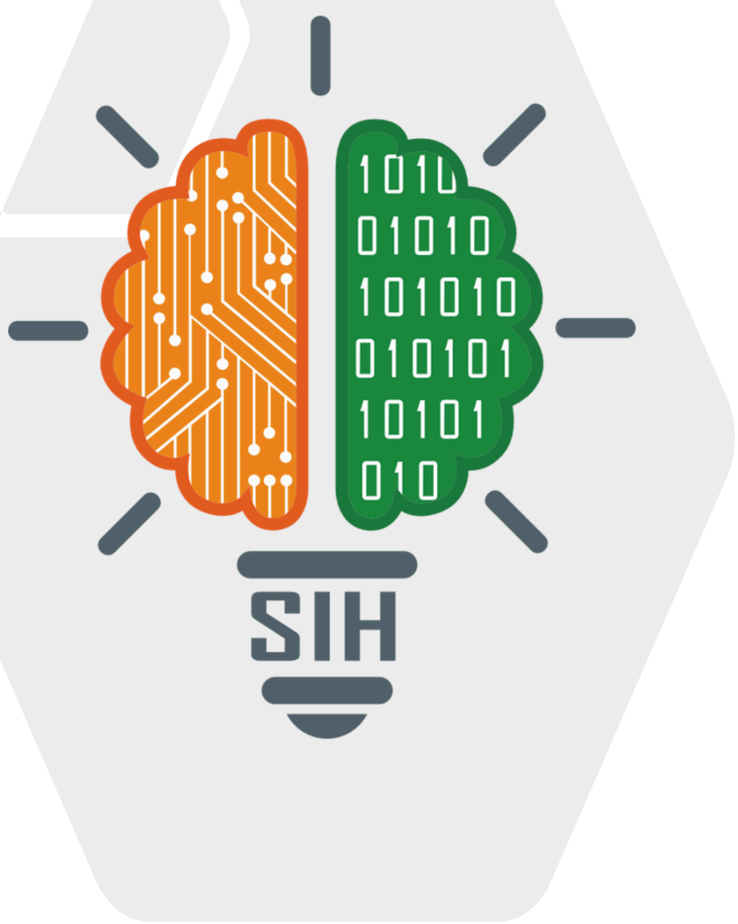
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TITLE PAGE

- **Problem Statement ID – SIH26121**
- **Problem Statement Title -** eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID - 177109**
- **Team Name - AnDas Apna Apna**



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IDEA TITLE

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IDEA & SOLUTION

NWIS (Nearby Wells Intelligence System) is an **AI-powered decision-support platform alongside eRTMAC**, bringing offset-well experience to the depth being drilled now.

- **AI document intelligence: OCR + NLP** turn decades of WCRs, DDRs and scans into a **searchable well knowledge base.**
- **Geospatial offset-well map:** wells within a chosen radius, with **losses, kicks, stuck pipe and NPT** marked.
- **Cross-well correlation:** compares wells **by depth and formation** and ranks the most similar offset wells.
- **Real-time eRTMAC integration:** matches **live depth and drilling parameters** against offset-well history.
- **Predictive risk forecasting: ML + anomaly detection** give **early warnings before risky depths,** with risk score and confidence.
- **Explainable AI + MCP assistant:** alerts and answers come from OIL's records via **MCP tools, each showing why, with source links.**

PROBLEM RESOLUTION

Problem: Drilling knowledge is **scattered across reports, scans and experts' memory.** Finding it takes **hours**, so **past problems repeat.** Records are **inconsistent**, rig networks are **weak**, and data is **too sensitive for public AI.**

Solution: NWIS **builds a searchable knowledge base automatically** and **warns before risky depths**, with **source-linked evidence.** It **standardizes old data, works offline at the rig** and **runs inside OIL's network.**

UNIQUENESS

Existing tools cover **one piece each**: documents, live dashboards, maps or chatbots. NWIS **combines them around the active well**, comparing **live data with OIL's history** and **warning before known problems, with evidence**. Its assistant uses **read-only MCP tools**, so **every answer comes from OIL's records**. It runs **inside OIL's network** and accepts **voice questions in Hindi, Assamese and English**.

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TECHNICAL APPROACH

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BUILDING THE INSTITUTIONAL MEMORY (one-time + continuous ingestion)





WCRs · DDRs · Mud Logs
Historical reports and scanned PDFs are ingested and extracted using OCR



NLP/AI Extraction
NLP pulls out depths, formations, events and actions



Structured Event Record
Extracted facts are structured and linked together



Well Knowledge Repository
Linked facts are stored as searchable memory each with its source page



eRTMAC Live Feed
Live drilling data is read from active wells

REAL-TIME DECISION SUPPORT AT THE RIG



AI Assistant (MCP)

Engineers ask questions by text or voice and get cited answers



Nearby Wells Finder

Offset wells are found on map and ranked by similarity



Correlation + Risk Engine

Live data is matched with history to score upcoming risks

FRAMEWORKS & TECHNOLOGIES

- **PaddleOCR / Tesseract:** convert scanned WCRs and DDRs into text.
- **PyMuPDF / pdfplumber:** pull text and tables from digital PDFs.
- **spaCy + Transformer NER:** extract depths, formations, events and actions.
- **PostgreSQL + PostGIS + pgvector:** store well records, run radius searches on the map, and power semantic search.
- **scikit-learn / XGBoost:** rank similar wells, detect anomalies and score risks.
- **Local LLM (Llama / Qwen via Ollama) + MCP:** cited answers with no data leaving OIL.
- **Whisper + IndicTrans2:** voice questions in Hindi and Assamese.
- **FastAPI + React (PWA) + Leaflet:** APIs, an offline-capable dashboard and the well map.

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FEASIBILITY AND VIABILITY

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Feasiblity

What makes this idea realistic is that:

- Drilling reports follow **recognisable structures (WCR/DDR)**, enabling **consistent data extraction**.
- **Open-source OCR, NLP, embeddings and local LLMs** keep the solution **cost-effective** and fully **on-premise**.
- **Read-only integration** with eRTMAC means **no disruption** to live operations.
- Our **modular pipeline (Documents → Repository → Search → Risk → Alerts)** can be built incrementally and validated at each stage.

Potential Challenges/Risks

- **Poor data quality and OCR errors** in historical scans.
- **Sparse incident data** for training reliable ML models.
- **False alerts** causing alert fatigue and loss of trust.
- **Limited real-time access** or infrastructure (no GPU, limited servers).
- **Over-reliance on AI** and user adoption in the field.

Proposed Solution

- **OCR quality scoring, range checks**, confidence scores and a human review queue for low-confidence pages.
- Start with **occurrence rates and rules**; add ML per risk as data grows, always **showing confidence**.
- **Severity tiers, multi-signal triggers**, de-duplication and feedback-based tuning.
- **Batch/file fallbacks**, manual depth entry, and **small CPU-friendly local models**.
- **Human-in-the-loop design**: the AI advises, qualified engineers decide, with explicit coverage warnings.

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IMPACT AND BENEFITS

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POTENTIAL AND IMPACT OF THE PROJECT

- **Cuts offset-well research from hours to seconds**, turning legacy reports into instant intelligence.
- Provides **early detection of drilling hazards** (kicks, losses, stuck pipe) before risky depths.
- Ensures **explainable, trusted decision-making** by linking every alert directly to source reports.
- Helps OIL crews **pre-empt formation risks and mitigations**, drastically reducing avoidable NPT.
- Creates an **enterprise-grade, secure AI framework** adaptable across all OIL drilling assets.

ऑयल इंडिया
OIL INDIA

BENEFITS OF THE SOLUTION

- **Social benefit**: Enhances rig safety; empowers **100% of field personnel** via voice AI in **2 regional languages (Hindi & Assamese)**.
- **Economic benefit**: Cuts avoidable NPT by **15–25%**, saving **₹15–30L+/day** in rig downtime and preventing costly fishing jobs.
- **Environmental benefit**: Targets **0 blowout incidents** and mitigates heavy mud losses (**>100 bbls**), safeguarding local groundwater.
- **Financial benefit**: Cuts research time by **>90% (<10s query turnaround)** with **100% on-premise data sovereignty** (0% cloud leak).

RESEARCH AND REFERENCES

DOCUMENTATION

Energistics WITSML Standard Documentation

DGH India – National Data Repository (NDR) Guidelines

Model Context Protocol (MCP) Documentation

vLLM: High-Throughput & Memory-Efficient LLM Serving

PaddleOCR Library Documentation

Qdrant Vector Database Documentation

RESEARCH PAPERS

Stuck-Pipe Prediction by Use of Automated Real-Time Modeling (SPE, 2014)

Retrieval-Augmented Generation for Knowledge-Intensive NLP (Lewis et al., 2020)

LayoutLM: Document Image Understanding and Information Extraction (Xu et al., 2020)

IndicTrans2: Accessible Multilingual Translation for Indian Languages (AI4Bharat, 2023)

THANK YOU!